

UNIT-I

FOOD

Food is any substance consumed to provide energy, growth, repair, and regulation of body processes. It is essential for maintaining health and preventing disease.

Food and its relation to health

The scientific study of food and its effects on the human body is a fundamental aspect of understanding overall health. Every bite we take influences our physiological processes, from cellular function to long-term well-being. It provides the evidence-based insights necessary to make informed decisions about our diets for health promotion.

Nutritional science is an interdisciplinary field that explores the relationship between diet and health. It investigates how the body processes and utilizes nutrients from food, examining their physiological roles.

Key food components

Food contains various components that nutritional science categorizes and studies for their specific roles in the body. Macronutrients are those needed in large quantities, providing energy and supporting numerous bodily functions. These include carbohydrates, proteins, and fats, each contributing differently to our physiological needs.

Carbohydrates are the body's primary and preferred source of fuel, easily converted into glucose for immediate energy by the brain, muscles, and other cells. Proteins provide amino acids, which are the building blocks for muscles, skin, blood, and structures within the brain and nervous system. While protein can supply energy, its main role is in cell building and repair, particularly after physical activity. Fats are also crucial for brain development, insulation, energy reserves, and protecting organs, providing the most concentrated source of calories per gram.

Micronutrients, comprising vitamins and minerals, are required in much smaller amounts but are equally important for health. Vitamins, organic compounds from plants and animals, are involved in energy production, immune function, and blood clotting. Minerals, inorganic elements from soil and water, play roles in growth, bone health, fluid balance, and various other bodily processes. Water and fiber also

contribute to overall health, with water being an essential medium for all bodily functions and fiber supporting digestive health.

In short

Carbohydrates

- Main source of energy
- Provide quick energy to the body
- **Sources:** rice, wheat, bread, potatoes, sugar

Proteins

- Needed for growth, repair, and maintenance of tissues
- Help in formation of enzymes and hormones
- **Sources:** milk, pulses, eggs, meat, fish, legumes

Fats

- Concentrated source of energy
- Help in absorption of fat-soluble vitamins (A, D, E, K)
- **Sources:** oils, butter, ghee, nuts, seeds

Vitamins

- Required in small amounts for normal metabolism
- Protect against deficiency diseases
- **Examples:** Vitamin A, B-complex, C, D, E, K

Minerals

- Essential for body structure and regulation
- **Examples:** calcium, iron, iodine, phosphorus, sodium

Water

- Helps in digestion, absorption, and transport of nutrients
- Regulates body temperature

Dietary Fibre

- Aids digestion and bowel movement
- Prevents constipation
- **Sources:** whole grains, fruits, vegetables

How Nutrition Influences Health

The components of food influence human health through intricate mechanisms, beginning with their absorption and utilization by the body. Once digested, nutrients are transported to cells, where they participate in various metabolic processes that sustain life. The impact of nutrition extends to cellular function, where specific nutrients are required for cellular growth, repair, and signaling pathways.

Nutrients affect the immune response, with certain vitamins and minerals, such as **vitamins C and D, zinc, and iron, supporting immune cell development and function. Protein intake is also linked to the production of antibodies and the activation of immune cells.** Dietary fats, particularly omega-3 fatty acids, can have anti-inflammatory properties, while saturated and trans fats may promote inflammation.

The gut microbiome, a community of microorganisms residing in the digestive tract, is significantly influenced by diet and plays a role in immunity. Dietary fiber, for instance, is fermented by gut bacteria, producing short-chain fatty acids that support immune function and help maintain the intestinal barrier. Conversely, diets high in fat and sugar can lead to reduced microbial diversity and an increase in opportunistic pathogens, contributing to intestinal inflammation. Unhealthy dietary patterns, characterized by high intake of processed foods, added sugars, and saturated fats, are linked to an increased risk of chronic conditions such as heart disease, type 2 diabetes, and certain cancers. Plant-based dietary patterns, emphasizing fruits, vegetables, whole grains, and legumes, have been consistently associated with a reduced risk of these diseases, partly due to their high fiber and antioxidant content, which can improve insulin sensitivity and reduce inflammation.

FUNCTIONS OF FOOD

At its core, food serves as fuel for our bodies. The primary source of energy for humans is carbohydrates, particularly glucose, which the body breaks down through cellular respiration to produce ATP—the energy currency that powers every cellular function. This energy supports everything from basic metabolic processes to complex physical activities.

Energy provision

Food contains six major groups of nutrients: proteins, fats, carbohydrates, vitamins, minerals, and water. Together, they offer the body the energy it requires to function, measured in calories. Each macronutrient contributes differently to our energy needs. Carbohydrates provide quick, readily available energy, while fats offer more sustained fuel over longer periods. Proteins, though primarily used for building and repairing tissues, can also be converted to energy when needed.

When the amount of ATP available exceeds the body's requirements, the liver uses the excess ATP and glucose to produce glycogen, which is stored in the liver and skeletal muscle cells for later use. This storage mechanism is crucial for maintaining energy during periods between meals or during intense physical activity.

Growth and tissue repair

Beyond energy, food provides the raw materials for building and maintaining our bodies. Amino acids are the building blocks of proteins, and protein breakdown provides amino acids that are used for cellular function. This is particularly important during childhood, adolescence, and pregnancy when rapid growth occurs, but tissue repair continues throughout life.

Regulation of body processes

Vitamins are essential organic molecules required in small quantities for many enzymes to function, and for this reason they're considered co-enzymes. Absence or low levels of vitamins can have a dramatic effect on health. Minerals similarly play crucial roles in maintaining structure and regulation throughout the body. Water and dietary fiber help regulate digestion, maintain body temperature, and eliminate waste products.

ENERGY

Energy is the **capacity to do work or produce heat**.

In nutrition, energy refers to the ability of food to support body functions, physical activity, growth, and maintenance.

Importance of Energy in the Human Body

Energy is required for:

- Basal metabolic activities (breathing, circulation, cell maintenance)
- Physical activity
- Growth and tissue repair
- Maintenance of body temperature
- Digestion, absorption, and metabolism of food

FORMS OF ENERGY

- Energy exists in various forms in the human body and in foods.

1. Chemical Energy

- Energy **stored in chemical bonds** of nutrients.
- Present in **carbohydrates, fats, and proteins**.
- Released during **metabolism** through oxidation.
- Example: Glucose oxidation releases energy for cellular work.

2. Mechanical Energy

- Energy used for **movement and physical work**.
- Includes:
 - Muscle contraction
 - Walking, running, lifting
 - Heart pumping blood
- Derived from chemical energy of food.

3. Heat (Thermal) Energy

- Energy released during metabolic reactions.
- Helps in:
 - **Maintaining body temperature**
 - Keeping enzymatic reactions active

- Excess energy is lost as heat.

4. Electrical Energy

- Involved in **nerve impulse transmission**.
- Important for:
 - Brain function
 - Muscle contraction
 - Heart rhythm
- Example: Electrical activity in nerves and muscles.

5. Potential Energy

- Stored energy due to position or state.
- Example:
 - Energy stored in **ATP molecules**
 - Stored fat in adipose tissue

6. Kinetic Energy

- Energy of motion.
- Example:
 - Blood circulation
 - Movement of muscles

ENERGY TRANSFORMATION IN THE BODY

- Food energy → **Chemical energy**
- Chemical energy → **Mechanical energy + Heat energy**
- ATP acts as the **energy currency** of the cell.

Form of Energy	Function in Body	Example
Chemical	Stored energy	Glucose, fat
Mechanical	Movement	Muscle action
Heat	Body temperature	Metabolism
Electrical	Nerve impulses	Brain signals
Potential	Stored energy	ATP
Kinetic	Motion	Blood flow

UNITS OF MEASUREMENT

The energy that we derive from food is essential for our bodies to function properly. This energy is measured in units that quantify the amount of energy released when the food is metabolized. The most common unit of measurement for food energy is the calorie. A **calorie** is defined as the amount of energy required to raise the temperature of one kilogram of water by one degree Celsius. In the context of nutrition, calories are used to express the energy content of foods and beverages.

While calories are the most commonly used unit of measurement for food energy, there are other units used in different contexts, especially in scientific and international settings.

In the International System of Units (SI), the unit of energy is the **joule (J)**. One joule is the energy expended when a force of one newton is applied over a distance of one meter. In the context of food energy, joules can be used, and 1 kilocalorie is equivalent to approximately 4.184 kilojoules. The use of joules is more common in scientific and technical applications outside of nutrition.

Calculating Food Energy

The energy content of food is calculated based on the amounts of carbohydrates, proteins, and fats it contains. Each of these macronutrients provides a different amount of energy per gram.

Energy Yield of Macronutrients

- Carbohydrates provide approximately 4 kcal/g.
- Proteins provide approximately 4 kcal/g.
- Fats provide approximately 9 kcal/g.
- Alcohol provides approximately 7 kcal/g.

These values are used to calculate the total energy content of a food item. The calculation involves multiplying the weight of each macronutrient in the food by its respective energy yield per gram and then summing these values.

Importance of Accurate Food Energy Measurement

Accurate measurement of food energy is crucial for various reasons, including **health and nutrition planning**, **food labeling**, and **research in nutrition and dietetics**. Understanding the energy content of foods helps individuals make informed choices about their diet, which can impact their health, weight, and overall well-being.

Applications in Health and Nutrition

In the context of health and nutrition, accurate food energy measurement is vital for:

- **Diet planning:** For weight management, athletic performance, and managing health conditions such as diabetes.
- **Food production and labeling:** To provide consumers with accurate information about the energy content of foods.
- **Nutritional research:** To understand how different diets and nutrients affect human health.

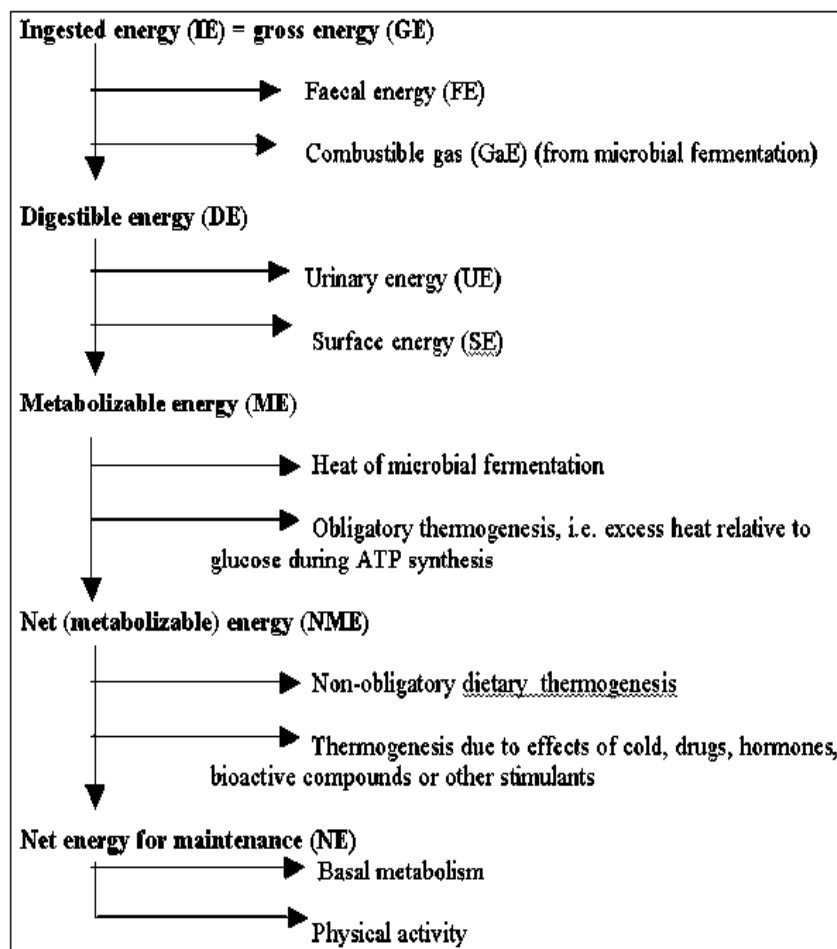
FLOW OF ENERGY THROUGH THE BODY

Food that is ingested contains energy - the maximum amount being reflected in the heat that is measured after complete combustion to carbon dioxide (CO₂) and water in a bomb calorimeter. This energy is referred to as ingested energy (IE) or gross energy (GE). Incomplete digestion of food in the small intestine, in some cases accompanied by fermentation of unabsorbed carbohydrate in the colon, results in losses of energy as faecal energy (FE) and so-called gaseous energy (GaE) in the form of combustible gases (e.g. hydrogen and methane). Short-chain (volatile) fatty acids are also formed in the process, some of which are absorbed and available as energy.

Most of the energy that is absorbed is available to human metabolism, but some is lost as urinary energy (UE), mainly in the form of nitrogenous waste compounds derived from incomplete catabolism of protein. A small amount of energy is also lost from the body surface (surface energy [SE]). The energy that remains after accounting for the important losses is known as “metabolizable energy” (ME).

Not all metabolizable energy is available for the production of ATP. Some energy is utilized during the metabolic processes associated with digestion, absorption and intermediary metabolism of food and can be measured as heat production; this is referred to as dietary-induced thermogenesis (DIT), or thermic effect of food, and varies with the type of food ingested.

This can be considered an obligatory energy expenditure and, theoretically, it can be related to the energy factors assigned to foods. When the energy lost to microbial fermentation and obligatory thermogenesis are subtracted from ME, the result is an expression of the energy content of food, which is referred to as net metabolizable energy (NME).



Physiological Fuel Values of Energy

Definition

Physiological fuel value refers to the **actual amount of energy (heat)** released in the human body when food nutrients are **digested, absorbed, and metabolized**.

(or)

Amount of energy available to the body from a given amount of nutrient is called as Physiological Fuel Value.

Four ways of specifying energy:

Gross energy (GE) is the energy of complete combustion measured as determined using bomb calorimetry.

Digestible energy (DE) is equal to the GE minus the energy remaining in fecal matter.

Metabolizable energy (ME) equals the GE minus the energy lost in fecal matter, urine, and combustible gases. We report ME with two procedures. Physiological fuel values are an estimate of ME and are calculated by assigning 4 kcal/gm of ME for protein, 4 kcal/gm for carbohydrates, and 9 kcal/gm for fat in the diet. A value is also given for ME based upon actual analyzed values for each species.

Net energy (NE) is the energy remaining for production after deducting from GE energy lost in the feces, urine, combustible gases, and body heat losses.

Gross food energy values are typically measured in a bomb calorimeter, which determines the energy in foods by measuring the heat released after complete burning of a food sample.

In the body, however, energy extracted from food is less than gross food energy value. As a result of incomplete digestion, absorption, and metabolism, a small amount of the energy value of the food is lost and excreted in the feces or the urine. Gross energy values can be adjusted to represent the energy actually available to the body. Such values are called **physiological fuel values** or energy nutrients.

Different nutrients yield different amounts of energy per gram due to:

1. Chemical structure

- Nutrients have different chemical compositions and molecular bonds.

- Fats have long hydrocarbon chains with many carbon–hydrogen bonds. (Highly reduced).
- Proteins contain amino acids with nitrogen groups. (C-H, N-H).
- Carbohydrates have simpler sugar units with fewer hydrogens per carbon. (C-H and O-H bonds)
- The **type and number of bonds** influence how much heat energy is released when bonds are broken during metabolism.

Therefore:

- **Fat > Carbohydrate > Protein** in energy density
- Because fats have more energy-rich bonds.

Energy depends on chemical structure because nutrients with more **reduced carbon (C–H bonds)** release more energy on oxidation (e.g., fats = 9 kcal/g), while those with more **oxygen or nitrogen** release less (carbs & proteins = 4 kcal/g).

2. Degree of oxidation

- The energy released during metabolism depends on how oxidized the nutrient already is.
- Nutrients that are **more reduced** (less oxygen, more hydrogen) release **more energy** upon oxidation.
- **Fats** are highly reduced → contain many C–H bonds → high energy (≈9 kcal/g).
- **Carbohydrates** are partially oxidized → fewer C–H bonds → moderate energy (≈4 kcal/g).
- **Proteins** yield less usable energy because oxidation also produces nitrogenous waste (urea).

In summary:

More reduced molecules = higher energy yield

Degree of oxidation affects energy yield because nutrients that are **highly reduced (more C–H bonds)** give more energy during oxidative metabolism, while **partially oxidized molecules (more C–O bonds)** give less. Hence fats yield 9 kcal/g while carbohydrates and proteins yield 4 kcal/g.

3. Biological availability

- Not all consumed energy is usable by the body.
- **Digestibility and absorption efficiency** affect energy availability.
- For example:
 - Dietary fiber in carbohydrates is poorly digested → contributes little/no energy.
 - Proteins have **metabolic losses** due to conversion of nitrogen into urea.
 - Fat is generally highly digestible and efficiently absorbed.
- Biological availability is influenced by:
 - ✓ food matrix (raw vs cooked, processed)
 - ✓ enzyme systems
 - ✓ gut health
 - ✓ antinutritional factors

Therefore, the **physiological fuel values (Atwater factors)** are lower than gross caloric values measured by bomb calorimeter.

Physiological Fuel Values (per gram)

Nutrient	Energy (kcal/g)	Energy (kJ/g)
Carbohydrates	4 kcal/g	17 kJ/g
Proteins	4 kcal/g	17 kJ/g
Fats	9 kcal/g	38 kJ/g
Alcohol (Ethanol)	7 kcal/g	29 kJ/g
Organic acids	3 kcal/g	13 kJ/g

Reason for Differences

- **Fats** produce the highest energy because they are more **reduced** and yield more ATP during oxidation.

- **Proteins** yield less usable energy due to:
 - loss of nitrogen (urea formation)
 - incomplete oxidation
- **Carbohydrates** are completely oxidized to $\text{CO}_2 + \text{H}_2\text{O}$.

DETERMINATION OF ENERGY VALUES OF FOODS

Introduction

- Foods provide energy to the body through carbohydrates, proteins, and fats.
- The energy content indicates how much heat is produced when food is oxidized in the body.
- It is commonly expressed in **kilocalories (kcal)** or **kilojoules (kJ)**

Principle

Energy value of food is determined based on:

1. Direct calorimetry

Direct calorimetry involves measuring the heat produced when a food sample is completely combusted in a controlled environment. This method provides a direct measurement of the energy content but is not commonly used for food analysis due to its complexity and the requirement for specialized equipment.

2. Indirect calorimetry

Indirect calorimetry, on the other hand, estimates the energy content of food based on its chemical composition.

Methods for Determination

1. Bomb Calorimeter Method (Direct Calorimetry)

The direct method for measuring the total energy content of food in a laboratory setting is called bomb calorimetry. This technique involves burning a dried food sample in a sealed, oxygen-rich container, known as a bomb calorimeter. The combustion releases heat, which is then absorbed by a surrounding water jacket.

The principle relies on measuring the temperature increase of the water. Since the heat capacity of water is known, the amount of heat released by the burning food can be precisely calculated. This method directly determines the gross energy

within the food sample. Bomb calorimetry is an accurate method for total energy measurement.

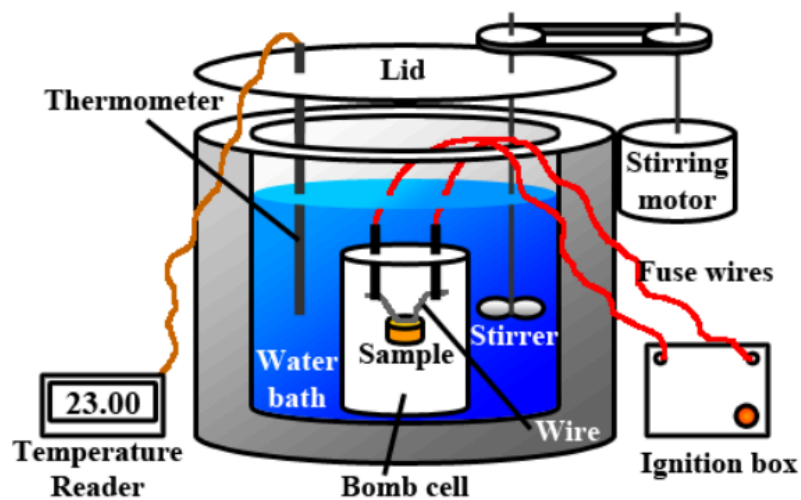
Key components

Combustion chamber (bomb)-where the food sample is burned.

Water jacket-surrounds the chamber to capture released heat.

High-pressure oxygen-ensure complete combustion.

Thermometer and sensors-measure water temperature changes with high precision.



Working of Bomb Calorimeter

Sample preparation- Foods are dried, homogenized and weighed. This step ensures consistent combustion and minimizes water interference.

Combustion process- The sample is sealed in the bomb, filled with oxygen and ignited. Complete combustion takes place in seconds.

Heat transfer- The generated heat rises the temperature of the surrounding water.

Data output- The system calculates energy content based on water's temperature increase, providing results in kcal/g or kJ/g.

2. Atwater system

The **Atwater system** is a widely used method for estimating the energy content of macronutrients in food. It relies on average caloric values for proteins, fats, and carbohydrates:

Represents energy actually available to the body after digestion and metabolism.

Standard Atwater factors:

- o **Carbohydrates:** 4 kcal/g (16.7 kJ/g)
- o **Proteins:** 4 kcal/g (16.7 kJ/g)
- o **Fats:** 9 kcal/g (37.6 kJ/g)
- o (Alcohol: 7 kcal/g)

Used in food labeling and dietary calculation

Steps in Determination of Energy Using Atwater Factors

1. Analyze food for **proximate composition**:
 - o Carbohydrate, Protein, Fat, Moisture, Ash
2. Multiply each macronutrient by its energy factor.
3. Add values to obtain total energy.

Example:

If a food contains:

- CHO = 10g $\rightarrow 10 \times 4 = 40$ kcal
 - Protein = 5g $\rightarrow 5 \times 4 = 20$ kcal
 - Fat = 2g $\rightarrow 2 \times 9 = 18$ kcal
- Total = 78 kcal

Modern Techniques and Technology

With advancements in technology, newer methods and instruments have been developed to measure the energy content of food more efficiently and accurately. These include **near-infrared spectroscopy (NIR)** and other spectroscopic

techniques that can rapidly analyze the chemical composition of food and estimate its energy content based on calibration models.

Factors Affecting Energy Values

- Digestibility of food (fiber reduces energy absorption)
- Cooking and processing
- Food composition (higher fat → higher energy)
- Age & health condition of individual

Significance

- Helps in diet planning and nutritional labeling
- Assists in assessing nutritional adequacy
- Helps understand caloric contribution of foods in metabolism

BASAL METABOLIC RATE

Basal Metabolic Rate (BMR) is the minimum amount of energy required by the body to maintain basic physiological functions such as breathing, circulation, cellular metabolism, and thermoregulation while at complete physical and mental rest, in a post-absorptive state, and at a thermoneutral environment.

(or)

BMR is the energy required to maintain basic vital functions of the body at rest, in a post-absorptive and thermoneutral state.

FACTORS AFFECTING BMR

Several factors influence your basal metabolic rate, including:

1. Body size. The more body tissue and cells you have, the more energy your body requires to maintain them.
2. Amount of lean muscle tissue. Muscle tissue requires a lot of energy to maintain itself.
3. Amount of adipose tissue (body fat). Compared to muscle tissue, fat cells require less energy to maintain themselves, but they require energy, nonetheless.

4. Sex. Males generally have a faster BMR because they're generally larger than females. Males also tend to have more lean muscle mass due to higher testosterone levels.
5. Age. Your BMR decreases with age mainly due to a loss of muscle mass. But hormonal and neurological changes can also affect your BMR as you age.
6. Race. Your race and/or ethnicity may affect your BMR. Researchers are still studying this.
7. Genetics. Your genes likely influence your BMR. Researchers are still studying how genetics affect metabolism.

Temporary situations that can affect your BMR include:

1. Skipping meals, fasting or starvation. Thanks to evolution, your body is always trying to protect itself from losing weight. If you consume too few calories, your body will slow down your BMR.
2. Environmental temperature: If you're in a very cold or very warm environment, your body must work harder to maintain a healthy body temperature (like through shivering or sweating). These scenarios increase your BMR.
3. Thyroid hormone levels. Elevated thyroid hormone levels (hyperthyroidism) increase your BMR. Low thyroid hormone levels (hypothyroidism) decrease your BMR.
4. Illness or injury. When you're sick or hurt, your BMR increases because your body is busy fighting off an infection and/or repairing tissues.
5. Stimulants. Stimulants like caffeine, nicotine and amphetamines can increase your BMR.